

- I. Explain the meaning of Version Control
 - A. Version Control is a system that records changes to files uploaded to a repository over time so you can recall specific versions later. The repository will track who made changes, when the changes were made and why the changes were made.
- II. Highlight a benefit of Version Control
 - A. A key benefit of version control is the ability to safely experiment without fear of inducing software bugs. With every change being recorded, if bugs are introduced in the syntax, the developer can simply download the latest good version and continue working.
- III. Provide an application of Version Control. This means, provide an example of how Version Control could be used.
 - A. Assume the scenario of teams of software developers working in different geographic regions, a company can reasonably expect the following cycle of development. Let's say there are three locations around the world, London, Los Angeles, and Mumbai. When the developers in London finish the tasks for the day, they upload to the repository. Then when the workers in Los Angeles come in, they can download the latest updates of the project London uploaded to

the repository. Likewise, Los Angeles will work on the project, uploading to the repository at the end of the day. When workers in Mumbai arrive at work, they can connect to the repository and download what the workers in Los Angeles completed. This speeds up the development cycle and ensures good well rounded programs. It also provides means for efficiencies and effective team work in developing a software project.

IV. Show a command used in Version Control (for example a Git command).

A. The following are GIT commands:

1. Git commit —amend (used to modify the most recent commit.)
2. Git diff (allows user to view the differences between your local unstaged changes and the latest version that you cloned or pulled.)